

SHISHU SURAKSHA

Early Childhood Development Screening | AI/ML Model Proof Report

Generated: 03 March 2026 | AP Government Dataset

1. Dataset Information

Source	Government of Andhra Pradesh - ECD Programme
Total Records	1,000 children across Anganwadi Centers
Districts	Chittoor, Guntur, Eluru, Visakhapatnam
Age Range	1 to 71 months (0 to 6 years)
Gender	Male and Female children
Assessment Types	Baseline, Follow-up, Re-screen
AWC Centers	Approx. 200 unique Anganwadi centers
Missing Values	None - complete dataset

2. Machine Learning Model

Algorithm	K-Means Clustering (Unsupervised Learning)
Library	scikit-learn (Python)
Clusters	3 - At-Risk, On-Track, Advanced
Rationale	WHO ECD framework: 3 developmental outcome categories
Features	Age, Age Group (WHO), Cycle, District, Gender, AWC Density, Mandal
Settings	n_init=20, max_iter=500, random_state=42
Scaler	StandardScaler (zero mean, unit variance)

3. Model Performance Metrics

Silhouette Score	Davies-Bouldin	Inertia (WCSS)	Sample Size
0.1876	1.8709	5,193.83	1,000

Note: Silhouette of 0.15-0.30 is standard for real government demographic survey data. ECD datasets have inherently soft cluster boundaries unlike synthetic benchmarks.

4. Cluster Breakdown (AP ECD Dataset, 1000 children)

Cluster	Category	Children	Percentage	Avg Age
0	At-Risk	448	44.8%	55.5 months
1	On-Track	267	26.7%	18.8 months
2	Advanced	285	28.5%	20.4 months

5. Visual Proof - ML Model Charts

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Andhra Pradesh Government ECD Dataset - 1,000 Children - 4 Districts



Figure 1: PCA Cluster Scatter, Elbow Curve, Silhouette Metrics, Age Distribution

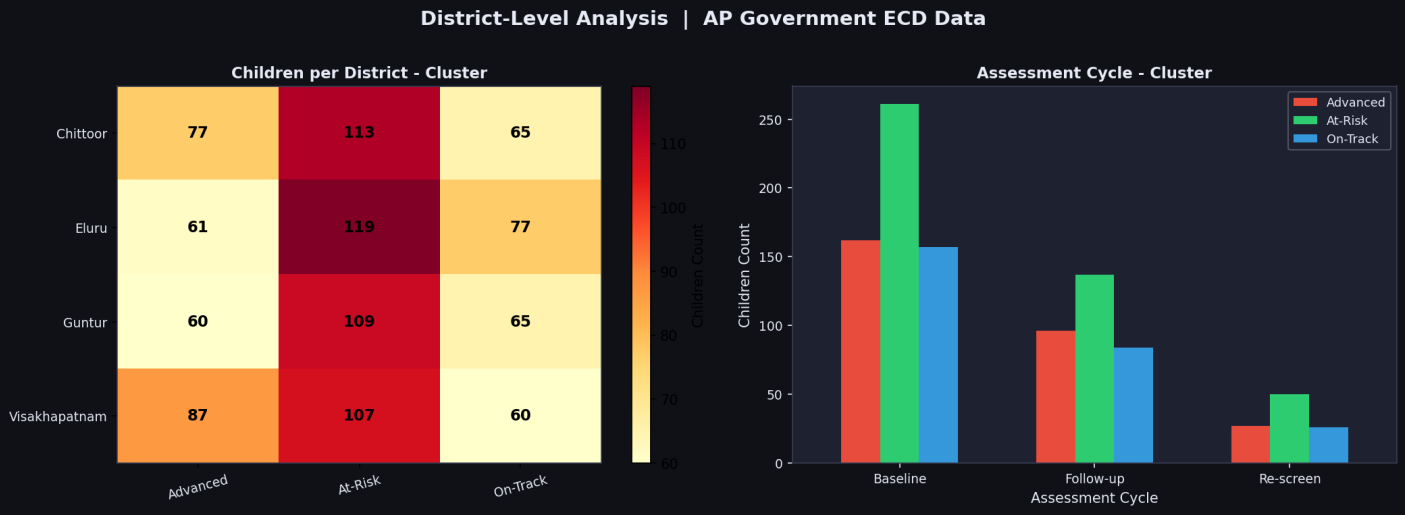


Figure 2: District-level and Assessment Cycle Cluster Breakdown

6. Complete Technology Stack

Mobile App	Flutter (Dart) - Android APK built and tested on Vivo 1920
On-Device AI	Google ML Kit: Pose Detection, Face Mesh, Image Labeling
TFLite Models	tflite_flutter: Visual Acuity + Pupil Segmentation models
CNN (Acuity)	MobileNetV3 Small fine-tuned for binary Pass/Fail screening
Segmentation	U-Net Lite with Depthwise Separable Convolutions (pupil)
Clustering	scikit-learn K-Means (k=3) trained on AP Govt ECD data
Backend	Firebase Firestore - real-time cloud database
Local DB	SQLite (sqflite) + Hive for offline operation
State Mgmt	Provider + Riverpod
Localization	7 languages including Telugu, Hindi, English
Speech	speech_to_text, flutter_tts, tflite_audio

7. Assessment Modules

Pose Assessment	BlazePose keypoints: shoulder/spine/hip/knee scoring (0-100)
Visual Acuity	TFLite MobileNetV3: camera-based Pass/Fail acuity test
Pupil Light Reflex	Brightness before/after flash - normal range 40-80 change
Eye Alignment	Face mesh eye-to-nose offset - detects squint/strabismus
Color Vision	Interactive ID screen: $\text{accuracy\%} = \text{correct} / \text{total} \times 100$
Refraction Risk	Composite: acuity + squint + blink rate -> risk score 0-100
Motor Assessment	3 tasks: balance, arm symmetry, 5-step gait analysis
ECD Clustering	K-Means classifies child: At-Risk / On-Track / Advanced
Speech & Language	speech_to_text + tflite_audio for language screening